

Problem 1

Let A , B , and C be random events. Explain the meaning of the equalities:

$$1^\circ \quad ABC = A; \quad 2^\circ \quad A + B + C = A$$

Solution

We will interpret these equalities in terms of set operations and event relationships.

Notation

In probability theory:

- ABC (or $A \cap B \cap C$) denotes the intersection: the event that A , B , and C all occur.
- $A + B + C$ (or $A \cup B \cup C$) denotes the union: the event that at least one of A , B , or C occurs.

Part 1°: $ABC = A$

Interpretation

The equality $ABC = A$ means:

$$A \cap B \cap C = A$$

This states that the event "all three events A , B , and C occur simultaneously" is equivalent to the event " A occurs."

What This Implies

For $A \cap B \cap C = A$ to hold, we need:

- Every outcome in A must also be in $B \cap C$
- Conversely, every outcome in $A \cap B \cap C$ must be in A (which is automatic)

Therefore: $A \subseteq B \cap C$

This means: $A \subseteq B$ and $A \subseteq C$

Meaning in Words

$$ABC = A \text{ means: } A \subseteq B \text{ and } A \subseteq C$$

In other words: **Whenever event A occurs, both events B and C must also occur.**

Equivalently: Event A implies both events B and C .

Example

Let:

- A = "It is snowing"
- B = "The temperature is at most 0°C "
- C = "There is precipitation"

Then $ABC = A$ because whenever it snows, we necessarily have both cold temperature and precipitation.

Part 2°: $A + B + C = A$

Interpretation

The equality $A + B + C = A$ means:

$$A \cup B \cup C = A$$

This states that the event "at least one of A , B , or C occurs" is equivalent to the event " A occurs."

What This Implies

For $A \cup B \cup C = A$ to hold, we need:

- Every outcome in $A \cup B \cup C$ must be in A
- Conversely, every outcome in A must be in $A \cup B \cup C$ (which is automatic)

Therefore: $B \cup C \subseteq A$

This means: $B \subseteq A$ and $C \subseteq A$

Meaning in Words

$$A + B + C = A \text{ means: } B \subseteq A \text{ and } C \subseteq A$$

In other words: **Both events B and C are subsets of event A . Whenever B or C occurs, A must occur.**

Equivalently: Events B and C both imply event A .

Example

Let:

- A = "The student passes the course"
- B = "The student gets an A grade"
- C = "The student gets a B grade"

Then $A + B + C = A$ because getting an A or getting a B both imply passing the course, and there are other ways to pass (getting a C, D, etc.).

Comparison

Equality	Set Relation	Meaning
$ABC = A$	$A \subseteq B \cap C$	A implies both B and C
$A + B + C = A$	$B \cup C \subseteq A$	Both B and C imply A

Final Answer

Part 1°:

$$ABC = A \iff A \subseteq B \cap C \iff (A \subseteq B \text{ and } A \subseteq C)$$

Meaning: Event A is a subset of both B and C . Whenever A occurs, both B and C must occur.

Part 2°:

$$A + B + C = A \iff B \cup C \subseteq A \iff (B \subseteq A \text{ and } C \subseteq A)$$

Meaning: Both events B and C are subsets of A . Whenever B or C occurs, A must occur.

Summary:

- $ABC = A$: A is more restrictive than both B and C
- $A + B + C = A$: A is more general than both B and C